

KISANKAVACH

Karnataka

QA, Testing, UAT & Requirements Traceability Matrix

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Farmer Credit. Benefits. Protection.

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Note: Page references are represented by section numbering to preserve stability across approved revisions.

1. Purpose and Scope

Defines the Karnataka pilot baseline for farmer-centred Kisan Credit Card enablement, benefit visibility, consent-driven data exchange, grievance handling, fraud protection and programme oversight.

In scope: farmer mobile experience, assisted service, participating-bank workflow, Government command centre, authorised integrations and operational reporting.

2. Guiding Principles

Kannada-first and low-literacy friendly; reuse verified information; explicit purpose-bound consent; least privilege; interoperability through adapters; configurable workflows and auditability.

The bank remains the final credit decision-maker. Government-originated and bank-originated data retain their source and provenance.

3. Quality Strategy

Risk-based testing covers functionality, integration, permissions, security, privacy, accessibility, performance, resilience, data migration and operations. Test evidence must be reproducible, environment-labelled and linked to build and requirement.

4. Test Levels

Level	Focus	Exit evidence
Unit/component	Rules, validators, adapters	Automated coverage and clean build
API/service	Contracts, state, idempotency, authorization	Contract suite and negative cases
System/E2E	Farmer, bank and Government journeys	Scenario evidence
Integration	FRUITS, banks, SMS/push, documents	Partner/simulator certification
Security/privacy	OWASP, access, consent, audit, leakage	No open critical/high risk
Performance/resilience	Load, soak, failover, recovery	SLO and recovery results
UAT	Business fitness and usability	Signed acceptance

5. Environment and Data

QA and UAT use synthetic or approved masked data. Test identities represent every role and geographic scope. External simulators support success, latency, timeout, malformed response, duplicate callback and partial failure.

6. Priority Test Scenarios

ID	Scenario	Expected result
TC-001	Kannada OTP login and masked mobile	Session created; safe display and audit
TC-002	Consent grant, retrieval and withdrawal	Access follows consent state and purpose
TC-003	Prefilled land/crop with provenance	Source and timestamp retained
TC-004	Draft save and resume on poor network	No loss or duplicate submission

ID	Scenario	Expected result
TC-005	Malicious document upload	Quarantined, blocked and security event
TC-006	Bank requests additional document	Farmer action and timeline update
TC-007	Sanction/rejection callback replay	Single valid transition
TC-008	District user requests other district	Denied and audited
TC-009	SLA breach	Escalation and dashboard reflected
TC-010	Dependency outage during submit	Safe retry and reconciled outcome

7. UAT Plan

UAT participants include farmer representatives, assisted operators, branch and regional bank users, programme officers and support teams. Sessions cover happy paths, exceptions, Kannada comprehension, accessibility, operational reports and incident runbooks.

8. Defect Management

Severity	Definition	Release rule
S1 Critical	Security exposure, data loss, system unavailable	Zero open
S2 High	Major journey blocked; no acceptable workaround	Zero open unless formal risk acceptance
S3 Medium	Impaired function with workaround	Owned plan and acceptance
S4 Low	Minor cosmetic/documentation issue	May defer by agreement

9. Requirements Traceability Matrix

Requirement	Design/API	Tests	UAT
BR-001 / FR-001	Identity, OTP endpoints	TC-001 + security	Farmer login
BR-002 / FR-002-003	Consent/Farmer services	TC-002-003	Consent clarity
BR-003 / FR-004-006	KCC/workflow	TC-004, TC-010	Application journey
BR-004 / FR-007	Document service	TC-005	Upload and bank request
BR-005 / FR-008-009	Timeline/SLA/events	TC-006-009	Status comprehension
BR-006 / FR-008	Bank portal/adaptor	TC-006-007	Officer workflow
BR-007 / FR-010-012	Grievance/reporting	Scope and SLA suite	Programme oversight

10. Entry, Exit and Sign-off

Entry: stable build, approved scope, test data, available environments and interfaces. Exit: planned execution complete, regression pass, performance and security approval, reconciled evidence, no prohibited open defects, operational readiness and signed UAT.

End of controlled baseline